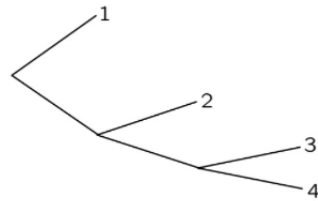
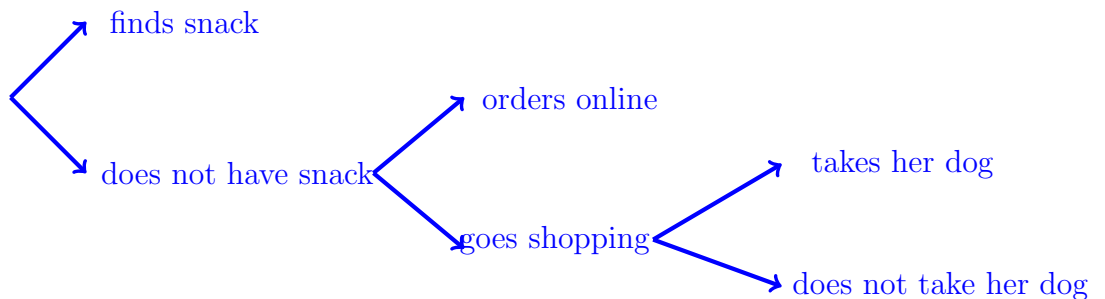


Homework 1- Solutions

Problem 1. Sarah checks the fridge to find a snack. If she finds her favourite snack, she will stay home. If she does not find her favourite snack, then She will either stay home and order online or she will go grocery shopping. If she goes for grocery shopping, she might either take her dog with her or leave her dog at home. In the diagram below which one shows that Sarah orders online.



- (a) 1
- (b) 3
- (c) 2
- (d) 4



Solution.

Based on the tree diagram the event that Sarah stays home and orders online correspond to leaf 2.

Problem 2. Suppose an experiment consists of tossing a two-sided coin four times. Consider the event $E = \{TTTT, HTTT, THTT, TTHT, TTTH\}$. Which of the followings describe E :

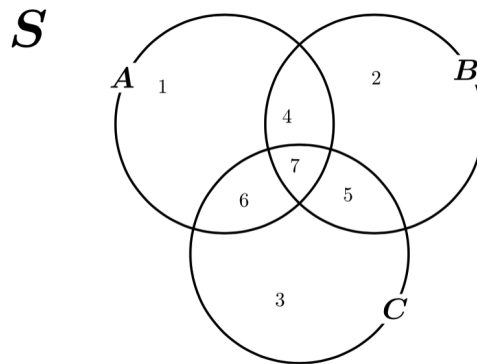
- (a) “At most once the coin turns heads”
- (b) “number of times tail occurs is more than two”

- (c) “At least three times the coin turns tails”
- (d) “number of tails is more than number of heads”

You may have more than one correct answer. Select every correct answer.

Solution. All are correct.

Problem 3. Sample space S as shown in the following figure is given. Which options are equivalent to the event $\{4, 6\}$. (Hint: Recall the set A minus B can be defined as $A - B = A \cap B'$)



- (a) $(B \cup C) \cap A$
- (b) $((B - C) \cup (C - B)) \cap A$
- (d) $(B \cap C' \cap A) \cup (B' \cap C \cap A)$

Solution. (b) and (d) are correct.

Problem 4. Let A and B be events on the same sample space, with $P(A) = 0.7$ and $P(B) = 0.5$. Can these two events be disjoint? Explain your answer.

Solution. It is False. Since if they are disjoint then $P(A \cap B) = 0$, hence, $P(A \cup B) = P(A) + P(B) = 1.2 > 1$, this means probability of event $A \cup B$ is bigger than one and that is impossible. Since no event can have probability greater than one.

Problem 5. Let A, B and C be pairwise disjoint subsets of the sample space. For each one of the following statements, determine whether it is true or false. Note: “False” means “It is not always true.”

- (a) $P(A^c) + P(B) \leq 1$

(b) $P(A) + P(B) \leq 1$

(c) $P(A \cup B \cup C) \geq P(A \cup C)$

Solution. (a) It is False. Since $P(A^c) = 1 - P(A)$. Hence, (a) is equivalent of saying $1 - P(A) + P(B) \leq 1$. That is equivalent of saying $P(B) \leq P(A)$. This is not always true. For example,

Let $A = \phi$, $B = S$, $C = \phi$, $P(A^c) + P(B) = 1 + 1 = 2$ or $P(B) = 1 > P(A) = 0$.

(b) It is true. $(A \cup B) \subset S$, hence $P(A \cup B) \leq P(S)$. $P(A \cup B) \leq 1$.

(c) It is true. $(A \cup C) \subseteq (A \cup B \cup C)$, hence $P(A \cup C) \leq P(A \cup B \cup C)$

Problem 6. A certain system can experience three different types of defects. Let A_i ($i= 1,2,3$) denote the event that the system has a defect of type i . Suppose that

$$P(A_1) = 0.12, P(A_2) = 0.07, P(A_3) = 0.05$$

$$P(A_1 \cup A_2) = 0.13, P(A_1 \cup A_3) = 0.14, P(A_2 \cup A_3) = 0.10,$$

$$P(A_1 \cap A_2 \cap A_3) = 0.01.$$

- What is the probability that the system does not have a type 2 defect?
- What is the probability that the system has both type 2 and type 3 defects?
- What is the probability that the system has both type 2 and type 3 defects but not a type 1 defect?
- What is the probability that the system has at most two of these defects?

(a). $P(\text{No } A_2) = 1 - P(A_2) = 1 - 0.07 = 0.93$

(b). We are looking for $P(A_2 \cap A_3)$ = probability of having both type 2 and type 3 defects.

$$P(A_2 \cap A_3) = P(A_2) + P(A_3) - P(A_2 \cup A_3) = 0.07 + 0.05 - 0.1 = 0.02$$

(c).

$$P(A_2 \cap A_3 \cap A_1^c) = P(A_2 \cap A_3) - P(A_1 \cap A_2 \cap A_3) = 0.02 - 0.01 = 0.01$$

(d) “At most two defects” means the system can have 0, 1, or 2 defects, but not all three.

$$P(\text{at most 2 defects}) = 1 - P(A_1 \cap A_2 \cap A_3) = 1 - 0.01 = 0.99$$

Problem 7. Let the sample space be the set of positive integers and suppose that $p(n) = \frac{2}{3^n}$, for $n = 1, 2, 3, \dots$. Find the probability of the set $\{5, 10, 15, \dots\}$, that is, the set of positive integers that are multiples of 5.

Solution. $A = \{5, 10, 15, \dots\}$.

$$P(A) = P(5) + P(10) + P(15) + P(20) + \dots$$

$$P(A) = \frac{2}{3^5} + \frac{2}{3^{10}} + \frac{2}{3^{15}} + \frac{2}{3^{20}} + \dots$$

$$P(A) = \frac{2}{3^5} \left(1 + \frac{1}{3^5} + \frac{1}{3^{10}} + \frac{1}{3^{15}} + \dots \right)$$

$$P(A) = \frac{2}{3^5} \left(1 + \frac{1}{243} + \frac{1}{243^2} + \frac{1}{243^3} + \dots \right)$$

$$P(A) = \frac{2}{243} \sum_{k=0}^{\infty} \left(\frac{1}{243} \right)^k$$

$$P(A) = \frac{2}{243} \left(\frac{1}{1 - \frac{1}{243}} \right) = \frac{2}{243} \left(\frac{1}{\frac{242}{243}} \right) = \frac{2}{242} = \frac{1}{121}$$

Problem 8. Out of the people in a seminar, 60% are mathematicians, 70% love coffee, and 48% fall into both categories. Determine the probability that a randomly selected person is neither a mathematicians nor a coffee lover.

The probability that a person is neither a mathematician nor a coffee lover is the complement of $P(M \cup C)$

$$P(M \cup C) = P(M) + P(C) - P(M \cap C) = 0.6 + 0.7 - 0.48 = 0.82$$

$$P(\text{neither}) = 1 - P(M \cup C) = 1 - 0.82 = 0.18$$